

Latency-Arbitrage

Sports Prediction Markets

Building an automated trading system that exploits the timing gap between live sports data feeds and broadcast television

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The Research Question

Prediction markets allow participants to trade contracts whose payoff depends on real-world events. When those events are live sporting matches, prices update in response to plays as they happen.

Do the participants pricing these contracts react fast enough? Can a computer with faster information consistently profit from the gap?

How Prediction Markets Work

A prediction market is a regulated exchange where contracts pay \$X if a future event occurs, and \$0 if it does not. Participants buy and sell these contracts at prices between 1¢ and \$X.

There are many prediction markets, but our focus is Kalshi. Kalshi is the largest CFTC-regulated venue in the United States, offering markets on a wide variety of events such as elections, economic indicators, weather, and live sporting events.

Additionally, Kalshi has a well documented SDK that allows for easy development and testing.

EXAMPLE CONTRACT

“Will the Lakers beat the Celtics?”

Current price:

65¢

→ market-implied probability: 65%

If Lakers win: contract pays \$1.00

If Lakers lose: contract pays \$0.00

Prices fluctuate continuously as the game unfolds.

Why Prices Move During a Game

Because the price of a contract is the market's estimate of probability, anything that changes the team's odds of winning will change the price.

PRE-GAME

65¢

Based on team strength, injuries, home advantage.

OPPONENT SCORES

58¢

The probability of winning drops; the price drops with it.

FINAL MINUTES, LEADING

92¢

A win is nearly certain; the price approaches \$1.00.

Information Travels at Different Speeds

Different participants in the market receive information about plays at very different times.

FAST	League data feeds NBA, NFL, MLB official APIs.	1–3 seconds
FAST	Sports media APIs ESPN, public scoreboard endpoints.	3–10 seconds
SLOW	Broadcast television Cable, streaming services.	30–60+ seconds

Roughly half a minute separates the slowest source from the fastest.

Trading the Information Gap

If a system reads from a fast data source, it can act before the slower-informed majority of participants.

Play occurs

An event happens on the field
(a score, a turnover, a key play)

System acts

Reads the API, estimates the
new fair price, places a trade at
the still-stale market price.

Market catches up

Television-watching
participants see the play and
trade. The price moves to its
new equilibrium.

By selling once the market has caught up, the system realizes the price difference as profit.

System Architecture

The pipeline runs continuously, repeating around twice per second across every live game.

1

DETECT

Poll live data feeds for active games and identify newly-occurred events.

2

ESTIMATE

Compute the expected new fair price using a model of how each event type affects win probability.

3

DECIDE

Compare the estimated fair price to the current market price; trade only if the gap exceeds transaction costs.

4

EXIT

Hold the position briefly; close it once the market has repriced or after a fixed timeout.

Reading the Game in Real Time

Continuous polling

The system queries four independent data sources every 500 milliseconds: ESPN, the NBA, the NFL, and Major League Baseball. Each query returns the current state of every live game: score, period, clock, possession.

Comparing each new snapshot to the previous one reveals what changed, which is the trading signal we act on.

Events the system reacts to

Scoring play

Any change in either team's score.

Turnover (NFL)

Possession changes without a scoring play.

Only the highest-confidence event types are included in this version. Complex signals such as injuries, ejections, weather are opportunities for future improvement.

Estimating the New Fair Price

When an event is detected, the system computes how much the market price should change. The model has two parts.

PART 1: BASE IMPACT

Empirical reference table

Drawn from published win-probability research in basketball, football, and baseball. Each entry maps an event type to a baseline change in win probability. For example, a late-game basket shifts the home team's probability by approximately 18 percentage points.

PART 2: CONTEXT ADJUSTMENT

Three multiplicative factors

- Magnitude: a three-pointer matters more than a free throw.
- Time remaining: late-game events have larger effects than early-game ones.
- Contract type: series-long contracts react less than single-game contracts.

From Play to Trade

Setup: Late in the fourth quarter, the away team scores a three-pointer. The market currently prices the home team's contract at 21¢.

Base impact (late-game scoring)	+18 percentage points
Magnitude (three-pointer)	× 1.4
Time-remaining factor (4:30 left)	× 0.91
Contract type (series-winner)	× 0.5

RESULT

The away team scored, so the home team's probability moves down: $-18 \times 1.4 \times 0.91 \times 0.5 \approx -11$ points

Estimated new fair price: 21¢ - 11¢ = 10¢

Market still showing 21¢ → an 11¢ gap. The system sells the home team's contract.

Trading is Not Free

Estimating the right direction is necessary but not sufficient. Every trade incurs a real cost: the difference between the price to buy and the price to sell at the same instant.

THE BID-ASK SPREAD

Buy at:

51¢

Sell at:

49¢

Round-trip cost:

2¢ per contract

A two-cent edge identified by the model is worth nothing, because the cost of executing both sides of the trade is two cents.

DECISION RULE

Trade only when expected profit clears the spread.

The system rejects any signal whose estimated edge (assuming worst-case fill prices on both entry and exit) is too small to be profitable. In practice this filters out the majority of detected events.

Phase One: Controlled Simulation

Before any live testing, the full pipeline was run against four scripted scenarios with synthetic price data and conservative fill assumptions.

Scenario	Direction	Price path	Result
NBA, late-game scoring play	Buy	50¢ → 63¢	+13¢
NFL, fourth-quarter turnover	Sell	51¢ → 58¢	+7¢
MLB, walk-off in extra innings	Buy	51¢ → 64¢	+13¢
NBA, series-winner contract	Buy	50¢ → 55¢	+5¢
Total			+38¢ across four contracts

All four scenarios closed profitably under the conservative cost assumptions. Simulation confirmed the pipeline's correctness, but a controlled environment is not evidence the live edge exists.

Phase Two: Live-Game Test

The first live deployment ran against an actual NBA playoff game using real Kalshi quotes but no real money.

Issues surfaced

Concurrent positions overwrote each other

When the system entered a second trade on the same contract, the first was silently lost. Roughly half of all opened positions vanished before being closed.

Model fired on low-impact plays

Routine free throws in garbage time were treated as full-strength signals. The contextual adjustments were not aggressive enough.

Network instability cascaded

A brief connection drop caused the polling loop to retry continuously, generating thousands of error logs and starving the trading loop.

Known Limitations & Future Work

Modeling assumptions

- Reference values come from public sports research, not from observed Kalshi prices. Future work: estimate them empirically from logged data.
- The contract-type adjustment is a single constant. A richer model would account for series state and games remaining.
- Multi-event sequences (e.g. a basket plus an immediate foul shot) are currently treated as one event.

Operational scope

- Only scoring plays and football turnovers are detected today. Injuries, ejections, weather, and lineup changes are out of scope.
- The system assumes any limit order at the prevailing ask fills immediately and fully. Thin order books would violate this in practice.
- All testing has been with simulated capital. Live trading would require additional safeguards before deployment.



Thank You